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Impact of Geochemical Reactions for Under-Ground Storage of Hydrogen Versus Ammonia

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Abstract

Challenges associated with Underground hydrogen storage (UHS), particularly those related to hydrogen's low density, high diffusivity, and the risk it poses on embrittlement of pipes, motivates consideration of ammonia as an energy carrier. This is because ammonia (NH3) is more compatible with pipeline materials and has a long history of industrial use with established regulations. Moreover, NH3 delivers energy at higher density compared to hydrogen (H2). All these benefits of ammonia can help speed up H2 transition for clean energy. This paper focuses on subsurface storage of H2 and NH3 as energy carriers.

Both H2 and NH3 are reactive fluids. It is crucial to model and understand the impact of geochemical reactions that may occur in underground environments at elevated pressures and temperatures. This study introduces a holistic approach to model reactive transport phenomena linked to hydrogen loss under geologic conditions. First, geochemical simulations with kinetics related to hydrogen loss in the subsurface are conducted. The results of both static and dynamic equilibrium reactions are then integrated into a multi-phase reactive transport flow simulator, providing a comprehensive simulation of hydrogen loss under realistic injection, storage, and production scenarios.

A homogeneous fully saturated aquifer with realistic mineralogical, petrophysical and fluid properties is assumed. Results indicate that sulfur (S) and iron(III) reduction are primarily responsible for the loss of H2 gas in the subsurface environment. Clay and silicate minerals exhibit minimal reactivity with H2 gas, suggesting that saline aquifers could be viable for hydrogen storage. Results also show that NH3 storage case demonstrates reduced geochemical reactions, promoting its significance in UHS. This builds up the case for NH3 as an alternate hydrogen carrier, considering the other perspectives of energy content, transportation cost, physical trapping efficiency, and operational safety.

This study provides valuable insights into the behavior of both H2 and NH3 in saline aquifers, advancing the field of safe and efficient underground hydrogen storage for large-scale energy applications.

Introduction

Hydrogen is a key component in the shift towards sustainable energy because of its energy density that is higher than that of natural gas by weight, and much lower carbon emissions throughout its lifecycle [1]. It is estimated that hydrogen production will have growth up to 10% [2]. Therefore, developing effective storage methods is critical in this transition to net-zero emissions. Hydrogen storage at surface facilities is challenging because of its low density, associated energy cost for compression and liquification, safety concerns, and the limited capacities of tanks [3]. Moreover, the seasonal nature of the energy demand requires storing hydrogen in large volumes available in underground formations, such as depleted reservoirs or saline aquifers, as an effective strategy for long-term storage [4,5]. For current small applications, hydrogen is stored in surface tanks or in salt caverns, which are relatively inert and impermeable, but are geographically limited [6].

UHS holds many potential benefits over caverns in terms of geographic availability, storage capacity and potentially lower CAPEX. However, low density, high diffusivity and the risk of geochemical and microbial reactions may result in significant hydrogen consumption, contamination, or loss [7–11]. Low density and high mobility of hydrogen favors first a vertical migration from the injection point, followed by lateral migration and accumulation near the caprock or near the spill point. Given the small size of its molecules, hydrogen may be able to penetrate and migrate through the caprock. Furthermore, its high diffusivity can lead to capillary trapping causing losses in storage efficiency. The chemical and biochemical reactions of hydrogen in the reservoir are additional concerns for the loss and purity in the production stream. In addition to subsurface storage challenges, hydrogen has a high cost of liquefaction and transportation. Since natural hydrogen resource plays are very few, economic hydrogen production is focused on production options from natural gas (blue hydrogen) which requires carbon capture and sequestration technology during its production.

To overcome some of these challenges, there has been some research looking into ammonia as a hydrogen carrier for sustainable energy storage and transport [12]. Ammonia generally offers higher hydrogen density than liquid hydrogen per unit volume (energy density of 5 kWh kg−1, a high hydrogen capacity of 17.8 wt%), which makes it a more attractive hydrogen (therefore energy) storage material [13]. Furthermore, ammonia has been commercially produced for over 100 years and current large-scale ammonia production is very stable [14]. The ease of liquefying ammonia either by compression to 1 MPa at room temperature (298 K) or by chilling to 240 K at ambient pressure (0.1 MPa) allows its storage, transportation, and distribution more energy-effective and cost-affordable than pure hydrogen. This significantly reduces its transportation costs and its associated development cost [15]. For example, shipping and pipeline infrastructure and transportation networks for ammonia over long distances have been already in place in many countries around the world. Ammonia has shown itself to be a promising alternative fuel in the shipping sector. The shipping sector transports more than 80% of the world trade by volume [16] and contributes to the 6th highest carbon emissions in the world. Consequently, making significant carbon cuts in the shipping sector emissions is of pressing importance [17, 18, 19, 20]. In addition, ammonia-related safety is an established industrial practice in the shipping sector.

Moreover, ammonia also shows its flexibility in terms of potential applications as an energy vector. Ammonia can be decomposed to supply CO and CO2 free hydrogen for a broad range of end uses, such as hydrogen fuel cell, or it can be used as a direct liquid fuel for electricity or power generation [21]. Despite the numerous advantages provided by ammonia usage, its major disadvantages generally include low burning velocity (SL) and high fuel nitrogen oxides (NOx) production [16]. Just like hydrogen, ammonia will also need capability for large volume storage potentially in underground formations to manage fluctuating seasonal demand.

There has been some ongoing research on understanding the processes involved for UHS, while very little focus is placed on understanding the underground ammonia storage (UAS) as a hydrogen career.

One important risk factor for both hydrogen and ammonia storage is the potential of geochemical and biochemical reactions [22-24] that can cause dissolution of minerals in the rock matrix, the formation of hydrogen sulfide (H2S) which can lead to sour corrosion [25], or potentially significant degradation and loss of stored volumes. Hydrogen sulfide can be generated abiotically [11, 26] since hydrogen has the potential to act as a reducing agent of sulfur species such as sulfate and pyrite. While sulfate reduction requires higher temperatures than the normative conditions typically encountered in saline aquifers, pyrite reduction can occur during the average deep saline aquifer temperatures.

The presence of geochemical reactions was a common observation during the storage of town gas in Europe, which led to the concerns that the same may happen for hydrogen storage implementations in saline aquifers or depleted gas reservoirs. Around twenty town gas geological storage projects were published by the technical association of the European natural gas industry [27]. Nearly half of these were aquifers and depleted gas reservoirs. Al Hamoud et al. [11] performed a reinterpretation of the published data and addressed the technical issues that were reported for these projects, specifically focusing on hydrogen loss and reactivity. All these projects experienced varying degrees of hydrogen loss and/or a change in gas composition.

A few recent publications [28,29,30] investigated different aspects of geochemical reactions of hydrogen in porous media. Zhan et al. [28] studied the influence of redox on the dissolution of minerals in UHS using PHREEQC software. They recommended excluding carbonate reservoirs for hydrogen storage and suggested clean sandstone formations for UHS. Saeed et al. [29] have similarly modeled the geochemical reactions of various minerals, also using PHREEQC. However, they focused on the impact of reactions on the porosity and permeability of the reservoir. They provided no discussion on the significance of each mineral in hydrogen loss. Rather, they focused on the impact of temperature, pressure, and salinity on hydrogen loss. They concluded that the abiotic reactions have minimum influence on hydrogen loss. Bo et al. [30] also performed geochemical modelling to examine the hydrogen loss associated with hydrogen dissolution and fluid-rock interactions using PHREEQC as a function of temperature and pressure. They investigated the potential hydrogen loss in two commercial gas storage reservoirs (Tubridgi and Mondarra) in Western Australia against the reservoir mineralogy, fluid properties, depth and temperature. They concluded that, unlike silicate and clay minerals, carbonates like calcite triggers up to 9.5% hydrogen loss due to calcite dissolution. Kinetic simulations show that Tubridgi only leads to 0.72% hydrogen loss, and Mondarra causes 2.76% hydrogen loss because of calcite dissolution and hydrogen dissociation in brines in a 30-year time. They suggested that deep calcite-free reservoirs together with calcite-free caprocks would be preferable for underground hydrogen storage.

The field of ammonia geo-storage is more recent and unexplored compared to hydrogen, facing much bigger uncertainties [31]. Just like for underground hydrogen storage, there are some important risk factors that need to be considered. The toxicity of ammonia can be a drawback for large-scale underground storage. Therefore, understanding the dissolution of ammonia in resident water and residual fluids is critical along with its migration risks away from the injection site. This must be addressed due to the risk of underground water and soil contamination. Dissolution of ammonia could also have significant impact on its purity and loss. It is also crucial to assess the reactivity of ammonia with rock minerals in the storage system.

It is usual to inject a cushion gas (such as CH4, N2, and CO2) for hydrogen storage in saline aquifers during the first year before the injection of working gas for pressure maintenance and creating an isolated region to reduce hydrogen loss [10]. The necessity of injecting a cushion fluid for subsurface ammonia storage can be similarly beneficial. It may even be needed to minimize dissolution of ammonia into in-situ fluids (for example, aquifer water). For most of the subsurface storage conditions, ammonia will be in a dense liquid phase. Therefore, using a dense cushion liquid (for example a cushion oil instead of a cushion gas) with a density greater than that of liquid ammonia can be injected prior to ammonia storage into underground storage space. A cushion oil injection can create a storage region while potentially minimizing the loss due to dissolution and geochemical reactions and mitigating HSE hazards. However, one may have to consider complex reactions and mass transfer between ammonia and cushion oil under subsurface conditions.

To the best of our knowledge, there is no published reactive transport modelling study on the possibility of underground ammonia storage in porous rock formations. The novelty of this work lies in its comparative field-scale analysis of geochemical reactions and their implications for storage of hydrogen versus ammonia in saline aquifers. This study integrates kinetic parameters from the PHREEQC database with advanced 3-D reactive transport modeling in PFLOTRAN to simulate chemical reaction kinetics and multiphase processes, revealing the mechanisms driving subsurface reactions and their influence on storage efficiency. By offering a detailed field-scale perspective, the findings provide a framework for advancing future direction of hydrogen storage considering carriers such as ammonia.

Methods

Ammonia as a Hydrogen Carrier

To ensure safe and cost-efficient underground hydrogen storage in potential geologic reservoirs, several key issues originating from gas-brine-mineral reactions, such as permeability change, formation of corrosive gases, and H_2 loss, are required to be thoroughly investigated. Despite its promising potential as an energy carrier, several unique properties of H_2 make it potentially more difficult to store in the subsurface than other intensively studied gases, namely CO_2 and CH_4. First, the low dynamic viscosity and density of H_2 compared to formation brine and/or denser cushion gases in the subsurface make it highly diffusive [25]. In addition, due to the highly reactive nature of H_2, hydrogen-driven redox reactions with subsurface minerals could not only lead to H_2 loss but also significant alterations in the sealing capacity and injectivity of the reservoir via mineral dissolution/precipitation. Furthermore, the toxic gases formed by the reactions, such as hydrogen sulfide (H_2S), could lead to potential health hazards in the storage sites. Therefore, understanding the reaction mechanisms of H_2 in underground storage conditions is crucial for accessing and developing storage sites.

As a potential alternative hydrogen carrier ammonia (NH_3) and its storage have become a topic of interest. NH_3 has long been utilized for producing fertilizers, chemicals, refrigerants, and fuels, so the demands and supply chain for NH_3 is significantly more developed than H_2. As illustrated in Table 1, the boiling point of NH_3 is considerably higher than that of H_2, resulting in a further cost reduction in liquefaction and transportation. From the perspective of Volumetric H_2 density, NH_3 is a better choice for H_2 storage. This is mainly attributed to the high density of NH_3. Besides the commercial benefits, the higher viscosity and the lower diffusivity of NH_3 compared to H_2 reduces the likelihood of hydrogen loss due to the gas diffusion through sealing rocks. From the risk management perspective, the distinct odor of NH_3 and its higher auto-ignition temperature makes it safer to handle and less flammable. Due to its unique characteristics in energy content, transportation cost, physical trapping efficiency, and operation safety, NH_3 and its underground storage should be investigated as another solution for the energy transition. This study aims to evaluate the underground storage and utilization efficiency of H_2 and NH_3 through modeling the gas-brine-mineral reactions under subsurface conditions.

Table 1—Summary of the properties of hydrogen and ammonia associated with underground storage efficiency and safety [25, 32, 33, 34].

Property (at 1 atm and 298.15K for the gaseous phase)	Hydrogen (H_2)	Ammonia (NH_3)
Dynamic viscosity (kg /m-s)	8.92×10^{-6}	1.01×10^{-5}
Density (kg /m^3)	70.8	682
Diffusivity in water (m^2/s)	5.13×10^{-9}	1.51×10^{-9}
Boiling point (K)	20.3	239.8
Volumetric energy density (MJ/L)	8 (Liquid)	12.7 (Liquid)
Volumetric H_2 density (kg-H_2/m^3)	70.9 (Liquid)	120.3 (Liquid)
Auto-ignition temperature (K)	844	924

Reactive Transport Modeling

Table 2 shows the mineral properties assumed for the storage site. Mineral reaction rates were calculated over a 12-month period to track changes in mineralogy and gas composition within the reservoir. Using surface area (SA), activity product (Q), equilibrium constant (Keq), and hydrogen ion activity (a_{H+}), mineral reaction rates (r) were determined through Equations 1 and 2:

$$r = k\left(1-\left(\frac{Q}{K_{eq}}\right)^p\right)^q SA \tag{1}$$

$$k = k_{298.15K}\exp\left(\frac{-E}{R}\left(\frac{1}{T}-\frac{1}{298.15}\right)\right)a^n{}_{H^+} \tag{2}$$

where E represents activation energy, $k_{298.15K}$is the dissolution rate constant at 298.15K, n indicates the reaction order for H^+, and R is the universal gas constant. Parameters p and q are empirical constants without dimensions, measured for only select minerals, potentially introducing uncertainty into rate calculations [35]. For conditions with uncertain or variable pH, Equation 3 shows the adjustments to the rate constant equation implemented in PHREEQC's kinetic rate functions. Palandri and Kharaka [35] specified different values for E, $k_{298.15K}$, and n under acidic, neutral, and basic conditions. It should be noted that PFLOTRAN's mineral kinetics module accepts only one rate constant specification.

$$\begin{aligned} k &= k_{298.15K} + k_{298.15K} + k_{298.15K} \\ &= k_{298.15K,\ acid}\exp\left(\frac{-E_{acid}}{R}\left(\frac{1}{T}-\frac{1}{298.15}\right)\right)a^{n_{acid}}{}_{H^+} \\ &+ k_{298.15K,\ neutral}\exp\left(\frac{-E_{neutral}}{R}\left(\frac{1}{T}-\frac{1}{298.15}\right)\right)a^{n_{neutral}}{}_{H^+} \\ &+ k_{298.15K,base}\exp\left(\frac{-E_{base}}{R}\left(\frac{1}{T}-\frac{1}{298.15}\right)\right)a^{n_{base}}{}_{H^+} \end{aligned} \tag{3}$$

When presenting the gas-brine-mineral reaction network, many previous studies have directly used the half-reactions (Tables 3-4) from PHREEQC's database. Despite being correct, the half-reactions are less intuitive because they involve free electrons (e^-) and do not contain H_2 directly in the reaction. In contrast, full reactions (Tables 5-6) clearly show the mole ratio and the final amount of products and reactants during the reaction and at equilibrium. Furthermore, the equilibrium constant of a full reaction involving hydrogen concisely indicates the tendency of the reaction under a hydrogen-rich environment. A full reaction is generated by summing the related half-reactions that detail the exchange of species in various phases. Applying the multiplicative properties of equilibrium constants, the $\log_{10}K_{eq}$ of a full reaction is the sum of the $\log_{10}K_{eq}$ of all the half-reactions involved.

Table 2—Summary of the mineral properties used for simulations. The dissolution rate and activation energy data are obtained from Palandri and Kharaka [35]. The equilibrium constants are obtained from the PFLOTRAN database (hanford.dat) [36].

Mineral	Volume%	Dissolution Rate at 25°C (mole/m²/s)	Activation Energy (kJ/mole)
Quartz (SiO_2)	78.2	4.571×10^{-14}	90.9
Kaolinite ($Al_2Si_2O_5(OH)_4$)	10.2	6.607×10^{-14}	22.2
Calcite ($CaCO_3$)	9.9	1.549×10^{-6}	23.5
Pyrite (FeS_2)	1.4	2.818×10^{-5}	56.9
Hematite (Fe_2O_3)	0.3	4.074×10^{-10}	66.2
Pyrrhotite (FeS)	0.0	9.120×10^{-9}	50.8
Magnetite (Fe_3O_4)	1×10^{-7}	2.570×10^{-9}	18.6

Table 3—Ammonia-brine-mineral reaction network modeled in the dynamic simulation with log10Keq values for various reactions at different temperatures [36]. Here, the laws of mass action are written in terms of O2 as the master redox variable, where equilibrium is assumed between O2 and H2, and the table shows the effective reactions for the basis swap of O2 to H2.

Half Reactions	Log K (0 °C)	Log K (25 °C)	Log K (60 °C)	Log K (100 °C)	Log K (150 °C)
$Fe^{3+} + 0.5\ H_2O \rightleftharpoons 0.25\ O_2(g) + Fe^{2+} + H^+$	-9.35	-7.77	-5.90	-4.15	-2.38
$Fe_2O_3 + 6\ H^+ \rightleftharpoons 2\ Fe^{3+} + 3\ H_2O$	2.14	0.11	-2.32	-4.61	-6.98
$Fe_3O_4 + 8\ H^+ \rightleftharpoons 2\ Fe^{3+} + Fe^{2+} + 4\ H_2O$	13.90	10.47	6.44	2.69	-1.12
$FeOOH + 3\ H^+ \rightleftharpoons Fe^{3+} + 2\ H_2O$	1.53	0.53	-0.60	-1.61	-2.60
$FeS + H^+ \rightleftharpoons Fe^{2+} + HS^-$	-3.64	-3.72	-3.92	-4.20	-4.62
$FeS_2 + H_2O \rightleftharpoons 0.25\ H^+ + 0.25\ SO_4^{2-} + Fe^{2+} + 1.75\ HS^-$	-26.50	-24.65	-22.75	-21.23	-20.02
$H_2S(g) \rightleftharpoons H^+ + HS^-$	-8.08	-7.98	-7.93	-7.96	-8.08
$NH_4^+ \rightleftharpoons H^+ + NH_3(aq)$	-10.07	-9.24	-8.28	-7.40	-6.52
$NH_4SO_4 \rightleftharpoons H^+ + NH_3(aq) + SO_4^{2-}$		-0.94			
$NH_3(g) \rightleftharpoons NH_3(aq)$	2.37	1.80	1.16	0.60	0.09
$N_2(aq) + 3\ H_2O \rightleftharpoons 1.5\ O_2(g) + 2\ NH_3(aq)$	-123.47	-112.11	-99.21	-87.57	-76.26
$NO_2^- + H^+ + H_2O \rightleftharpoons NH_3(aq) + 1.5\ O_2(g)$	-47.44	-42.52	-36.79	-31.48	-26.16
$NO_3^- + H^+ + H_2O \rightleftharpoons NH_3(aq) + 2\ O_2(g)$	-62.86	-56.30	-48.70	-41.68	-34.69

Table 4—Hydrogen-brine-mineral reaction network modeled in the dynamic simulation with log10Keq values for various reactions at different temperatures [36]. Here, the laws of mass action are written in terms of O2 as the master redox variable, where equilibrium is assumed between O2 and H2, and the table shows the effective reactions for the basis swap of O2 to H2.

Half Reactions	Log K (0 °C)	Log K (25 °C)	Log K (60 °C)	Log K (100 °C)	Log K (150 °C)
$CaCO_3 + H^+ \rightleftharpoons Ca^{2+} + HCO_3^-$	2.23	1.85	1.33	0.77	0.10
$CO_3^{2-} + H^+ \rightleftharpoons HCO_3^-$	10.62	10.33	10.13	10.08	10.20
$Al_2Si_2O_5(OH)_4 + 6H^+ \rightleftharpoons 2\ Al^{3+} + 2SiO_2(aq) + 5H_2O$	9.02	6.81	3.85	0.95	-2.02
$SiO_2 \rightleftharpoons SiO_2(aq)$	-4.63	-4.00	-3.47	-3.08	-2.72
$HSiO_3^- + H^+ \rightleftharpoons H_2O + SiO_2(aq)$	10.32	9.95	9.47	9.08	8.85
$CO_2(g) + H_2O \rightleftharpoons H^+ + HCO_3^-$	-7.68	-7.81	-8.05	-8.36	-8.77
$CO_2(aq) + H_2O \rightleftharpoons H^+ + HCO_3^-$	-6.58	-6.34	-6.27	-6.39	-6.72
$H_2(g) + 0.5O_2(g) \rightleftharpoons H_2O$	46.14	41.55	36.30	31.52	26.85
$H_2(aq) + 0.5O_2(g) \rightleftharpoons H_2O$	49.15	44.66	39.44	34.62	29.82

Table 5—Summary of full mineral buffer reactions for hydrogen based on reactions in Table 3. The equilibrium constants at 298 K are provided.

Full Reactions	Log K (25 °C)
$FeS_2 + H_2(aq) \rightleftharpoons FeS + H_2S(aq)$	-2.42
$FeS_2 + \frac{1}{3}Fe_3O_4 + \frac{4}{3}H_2(aq) \rightleftharpoons 2FeS + \frac{4}{3}H_2O$	7.51
$Fe_2O_3 + 2FeS_2 + 3H_2(aq) \rightleftharpoons 4FeS + 3H_2O$	17.86
$Fe_2O_3 + 3FeS_2 + \frac{1}{3}Fe_3O_4 + \frac{13}{3}H_2(aq) \rightleftharpoons 6FeS + \frac{13}{3}H_2O$	25.37

Table 6—Summary of full mineral buffer reactions for ammonia based on reactions in Table 4. The equilibrium constants at 298 K are provided.

Full Reactions	Log K (25 °C)
$6Fe^{3+} + NH_3(g) + 2H_2O \rightleftharpoons NO_2^- + 6Fe^{2+} + 7H^+$	-2.28
$8Fe^{3+} + NH_3(g) + 3H_2O \rightleftharpoons NO_3^- + 8Fe^{2+} + 9H^+$	-4.02
$6Fe^{3+} + 2NH_3(g) \rightleftharpoons N_2(aq) + 6Fe^{2+} + 6H^+$	67.32
$4FeS_2 + 4H_2O + NH_3(g) \rightleftharpoons 3H_2S(g) + 4FeS + NH_4SO_4 + H^+$	-57.07

Reservoir Information

The storage reservoir is modeled as a homogeneous formation measuring 2,000 m × 2,000 m × 200 m, divided into 156,250 grid blocks (each 16m × 16m × 20m). The injection well is centrally positioned in the X-Y plane and extends through the top 5 vertical grid blocks. We implemented a Peaceman well model [37] with consistent mass injection rate. During a 3-month period, approximately 1.925×106 kg (or 9.549×108 moles) of H_2 or 1.5×108 kg (or 8.807×109 moles) of NH3 gas was introduced into the aquifer, causing bottom hole pressure to increase from 10 to around 13 MPa. No-flow boundaries were applied on all sides of the reservoir model. The aquifer contains typical sandstone minerals, with storage parameters detailed in Table 7. Phase equilibria were characterized using the Redlich-Kwong-Soave (RKS) equation of state [38]. For transport modeling, we applied the van Genuchten-Mualem function (M=0.3) for $H_2(g)$ and water relative permeability. The aquifer's capillary pressure follows a van Genuchten function with M=0.5 and α = 1 × 10^{-4}. For the H2 storage case, gas diffusivity was established as 1 × 10^{-8} m^2/s. For the NH3 storage case, it is already in the liquid phase as indicated in Table 8.

Table 7—The storage conditions and properties of the aquifer used in this study.

Property	Value
Porosity	0.2
Permeability (mD)	200.0
Temperature (°C)	50.0
Initial pressure (MPa)	10.0
Initial brine pH	8.6
Initial brine salinity (ppm)	35,402.3
Geothermal gradient (K/m)	2.5 × 10^{-2}
Pressure gradient (kPa/m)	10.5

Table 8—The properties of hydrogen and ammonia under storage conditions obtained from the NIST database [41].

H-carrier	Temperature (C)	Pressure (MPa)	Density (kg/m^3)	Volume (m^3/kg)	Viscosity (µPa*s)	Phase
H_2	50	13	9.0804	0.11013	9.5602	supercritical
NH_3	50	13	576.78	0.0017338	109.92	liquid

Simulation Workflow

This study presents a comprehensive reaction network for underground hydrogen and ammonia storage, complete with kinetic rates and equilibrium constants that can be integrated into other reservoir simulation models capable of handling complex storage sites and transport phenomena. We propose a workflow that couples static and kinetic geochemical simulations from PHREEQC with multiphase flow simulations using PFLOTRAN [36], an advanced open-source computational framework for reservoir modeling. To more accurately represent real underground storage conditions, we developed dynamic models that incorporate both reaction kinetics and multiphase mass transport phenomena. The gas-brine-mineral system's static and kinetic equilibrium conditions calculated via PHREEQC provide essential initial and boundary chemical constraints for these dynamic models. This approach improves upon previous simulation studies that were typically limited to 1-D single-phase flow, which only considered aqueous diffusion of stored gas in mass transfer processes, potentially underestimating gas loss. On the other hand, such 1-D models often assumed homogeneous, well-mixed systems with instantaneous transport of reactants and products, potentially overestimating reaction extents. The effectiveness of our similar methodological approach has been previously demonstrated in previous reservoir engineering study [39].

Table 1 enumerates all minerals examined in this study. Each mineral's quantity (in moles) was determined from its density and volume fraction in the reservoir rock. This data was then utilized in PHREEQC phase equilibrium simulations across all scenarios. Due to the inclusion of several important minerals, we employed the llnl.dat database (developed by Livermore National Laboratory) with PHREEQC version 3.7.3 for all static and kinetic simulations [40]. Using PHREEQC's statically equilibrated brine composition as the initial condition, we observed minimal changes in brine chemistry over a 10-year PFLOTRAN simulation period without additional gas introduction. This stability confirms that PHREEQC's output effectively establishes an equilibrium for PFLOTRAN simulations. The static equilibrium simulation, without H_2(g) injection, showed that equilibrating all minerals with initial reservoir brine salinity of 35,402.3 ppm yielded an initial brine pH of approximately 8.6 and hydrogen molality around 1×10^{-8} mole/L across all scenarios.

At the injection well, rapid gas introduction created distinct chemical conditions compared to the broader reservoir. Since dynamic simulation requires fixed boundary chemical composition, understanding the chemical state near the well shortly after injection, rather than at equilibrium, is essential. PHREEQC's kinetic rate functions can adjust to changing pH conditions dynamically. This adaptability is crucial in environments with uncertain pH fluctuations, enabling the creation of boundary conditions that accommodate sudden reactant influxes. Using kinetic models, we tracked monthly variations in composition for all phases at the well. One year post-gas injection, brine pH stabilizes around 7.7 for the H2 case and 11.3 for the NH3 case, with H2 molality reaching approximately 8×10^{-2} mol/L for the H2 case and NH3 molality around 30 mol/L. Based on this near-steady-state result, well brine pH was standardized at 7.7 and 11.3 for the H2 and NH3 cases as a boundary condition, respectively.

Results and Discussions

Amount of Hydrogen Stored

Under the same reservoir pressure limit of 13 MPa, compared to using H2, using NH3 as the hydrogen carrier allows 16 times more hydrogen storage.

Table 9—The amount of hydrogen gas and ammonia injected and stored to raise the aquifer pressure to 13 MPa. Properties are obtained from the NIST database [41].

H-carrier	Amount Injected (kg)	specific molar content (mol/kg)	Amount Injected (mol)	Amount H Retained after 1 year (mol)
H_2	1.925×10^{6}	496.05	9.549×10^{8}	1.393×10^{9}
NH_3	1.5×10^{8}	58.719	9.549×10^{8}	2.636×10^{10}

Reactive Transport of NH_3 vs. H_2

Figs. 1 and 2 illustrate the hydrogen plume progression during the first year of UHS. For the hydrogen (H_2) storage scenario, clay, silicate, and carbonate minerals result in significantly less abiotic H_2(g) loss compared to Fe(III)-bearing minerals. Additionally, the reduction of Fe(III) minerals promotes the dissolution of Fe(II). Both reductive pyrite dissolution and Fe(III) reduction contribute to pyrrhotite precipitation. Although magnetite is also a key precipitate in the hydrogen–iron reaction network, it is rapidly reduced by dissolved H_2(aq) and the H_2S(aq) generated during these processes. Consequently, the H_2S(aq) produced from pyrite dissolution and pyrrhotite precipitation is consumed during Fe(III) reduction. These results suggest that sandstone-rich reservoirs are well-suited for minimizing H_2(g) loss due to geochemical reactions.

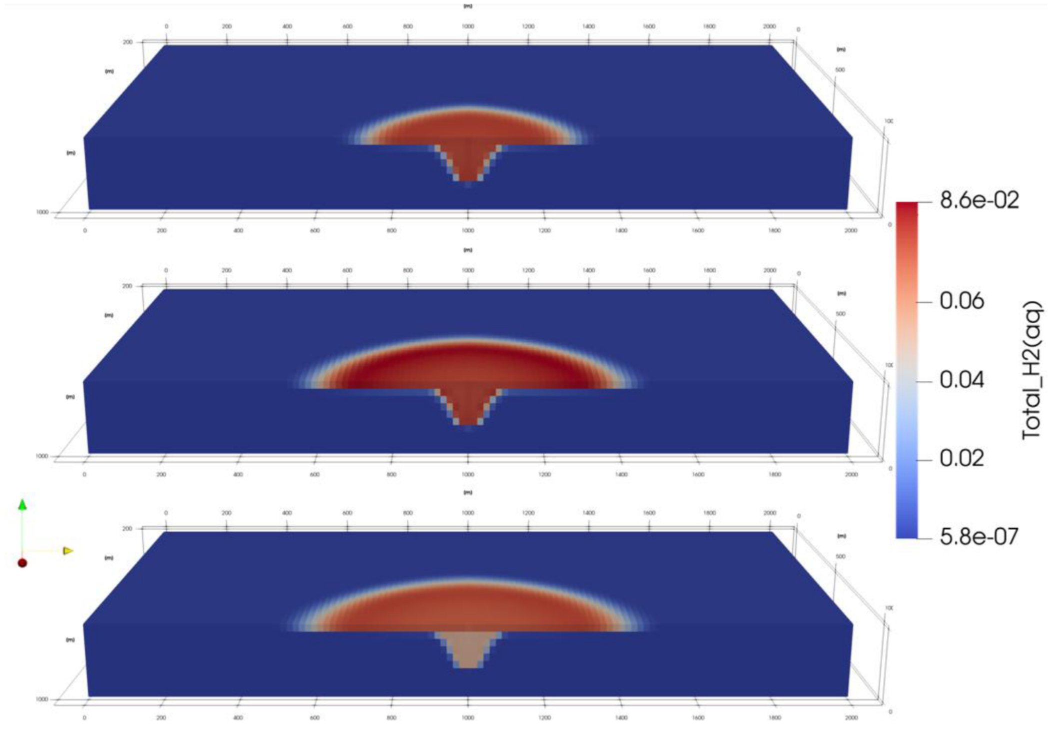

Figure 1—The molality of dissolved H2(aq) in the 1st (top), 3rd (middle), and 12th (bottom) month of storage.

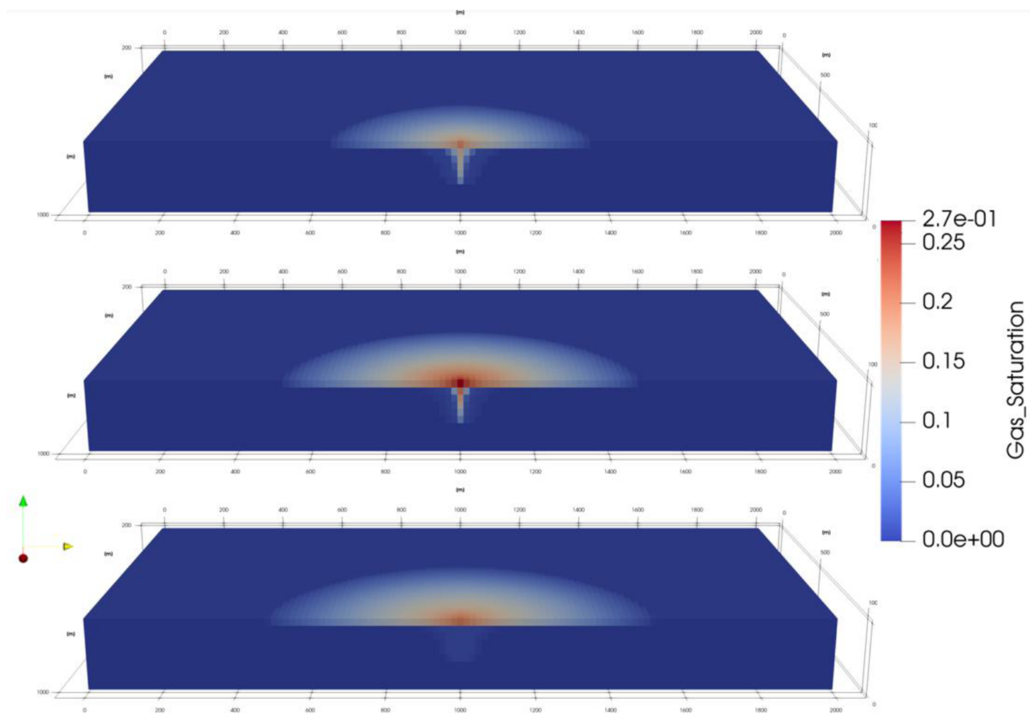

Figure 2—The gas saturation of the injected H2(g) in the 1st (top), 3rd (middle), and 12th (bottom) month of storage.

As illustrated in Fig.3, the overall porosity changes remain negligible due to abiotic reactions. This indicates the limited hydrog eochemical impact on storage capacity.

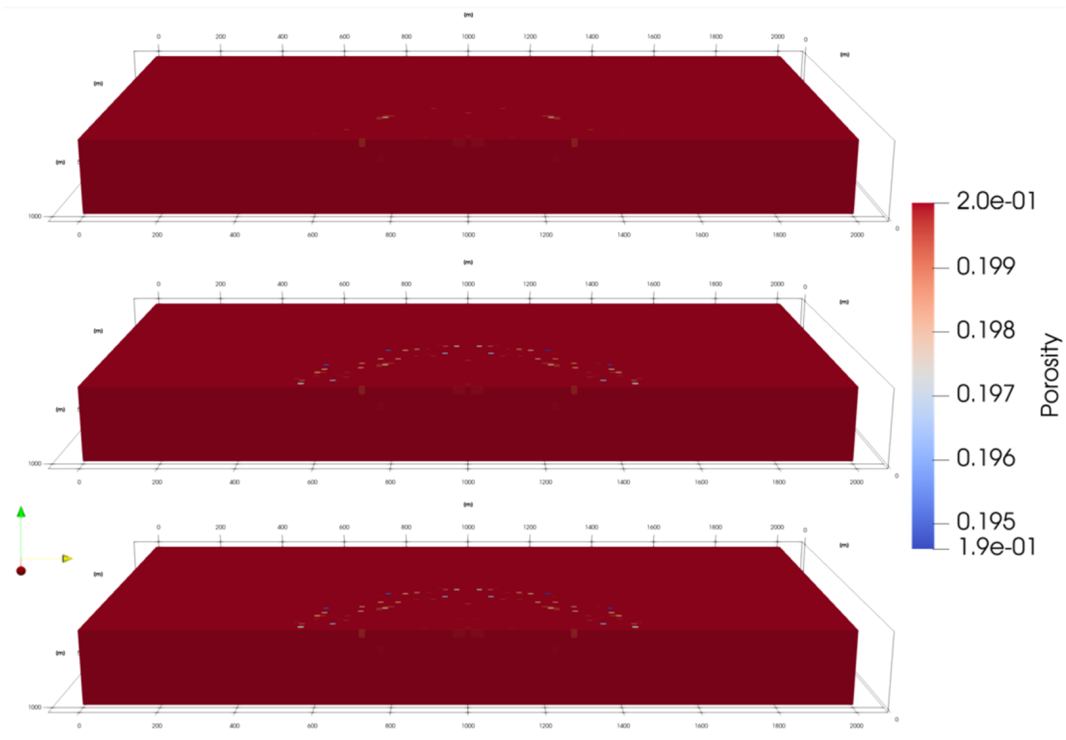

Figure 3—The porosity maps for the hydrogen storage case in the 1st (top), 3rd (middle), and 12th (bottom) month of storage.

Fig. 4 shows the injected NH3 progression during the first year. As expected, NH3 stays around the injection site and does not show migration into upper layers unlike H2 (Fig. 2). Figs. 5 and 6 show the major products from Ammonia (NH3) reaction in the subsurface. NH3 is a reducing agent in redox chemistry. Its nitrogen is already at the lowest common oxidation state (-3), so it can donate electrons and be oxidized to N_2 under mildly oxidizing conditions and participates in acid–base equilibrium with ammonium (NH4+). In contrast, forward oxidation of NH3 to nitrite (NO2-) or nitrate (NO3-) is thermodynamically unfavorable in abiotic subsurface environments. For ammonia storage, the main contributor to ammonia loss in the subsurface is the reduction of Fe(III)-bearing minerals, such as hematite, as shown in Table 6.

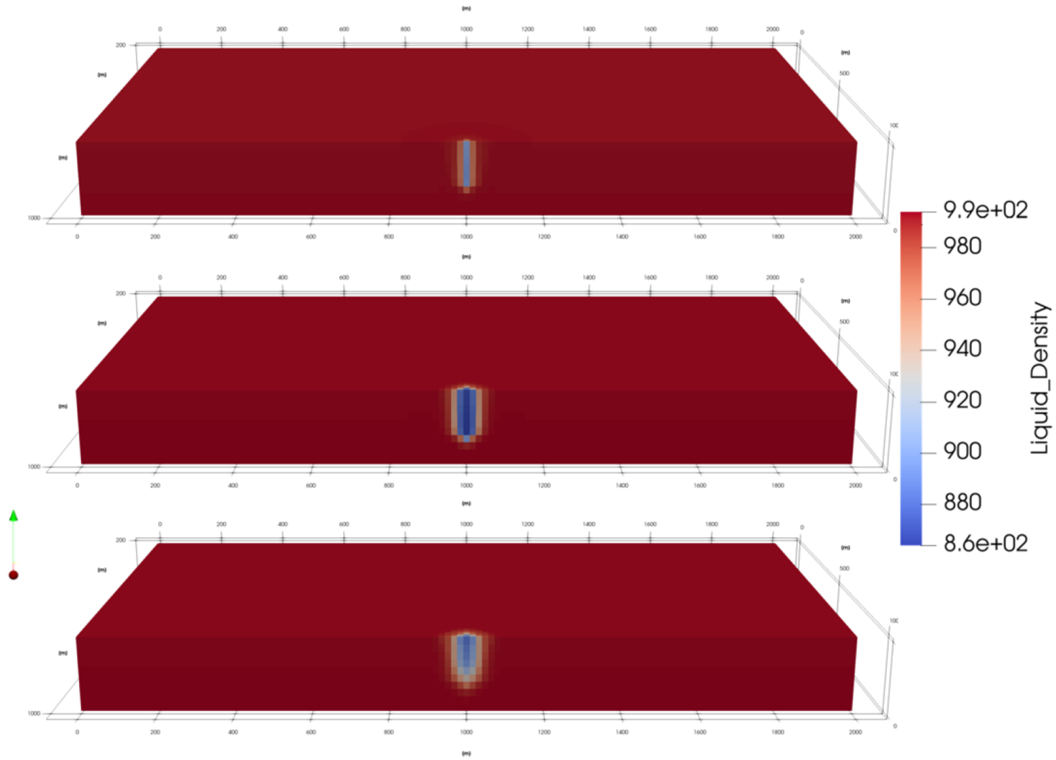

Figure 4—The liquid density for the ammonia storage case in the 1st (top), 3rd (middle), and 12th (bottom) month of storage.

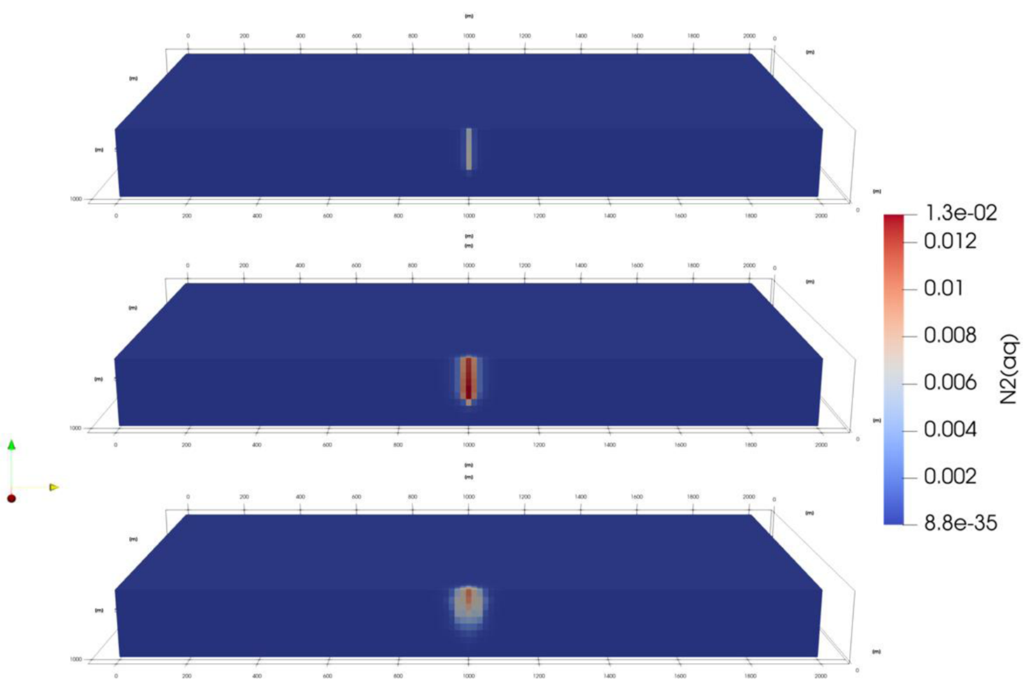

Figure 5—The molality of generated N2(aq) in the 1st (top), 3rd (middle), and 12th (bottom) month of storage.

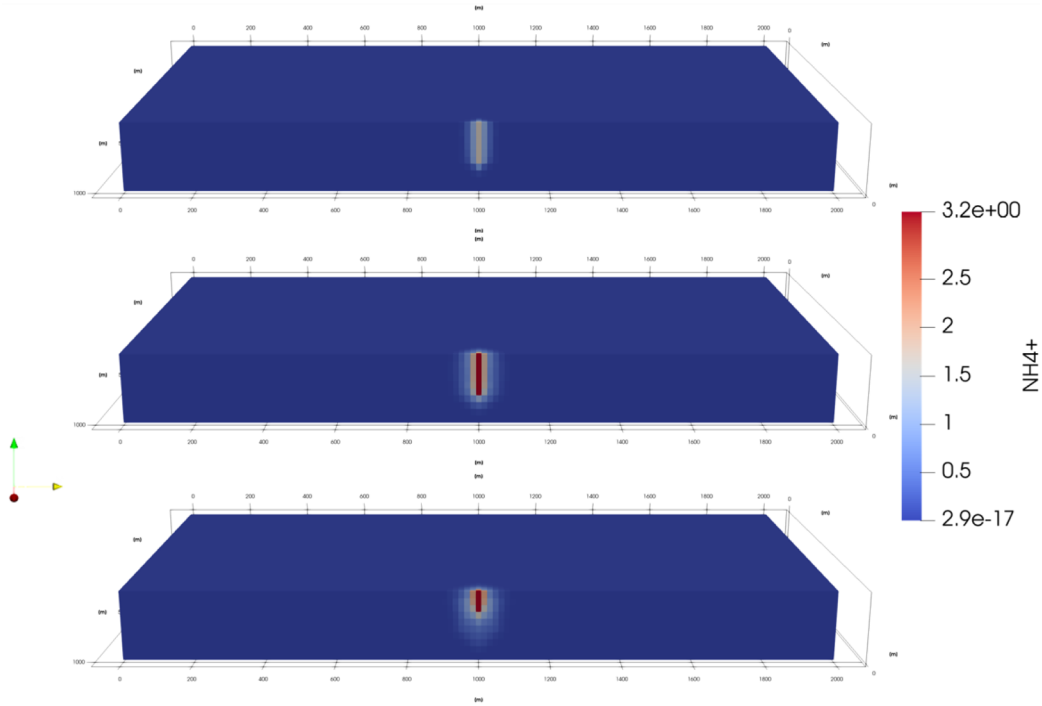

Figure 6—The molality of generated NH4+(aq) in the 1st (top), 3rd (middle), and 12th (bottom) month of storage.

Table 8 shows that NH3 should be in liquid phase under storage conditions. Due to the limitations in the EOS implementations for PFLOTRAN, we could not find the exact suitable EOS for NH3 under the storage conditions. Instead, we defined a relatively small Henry's constant (1×107 Pa) for NH3 to force its dissolution in the brine, and we manually set the density and viscosity for the dissolved gas phase. As shown in Fig. 4, the liquid density of the brine-ammonia mixture resulting from our implementation matched the expected properties.

Figs. 7 and 8 show the simulated gas and mineral variations over the first year of storage. On the reaction side, the loss is mainly due to Fe(III) and S reduction. In contrast, the injected NH3 is already in the liquid phase and exhibits substantially lower reactivity in the subsurface, resulting in greater retention. The primary reactive loss mechanism for NH3 is its reduction of Fe(III) to form N2, which accounts for only 0.2% loss of the total NH3 over one year.

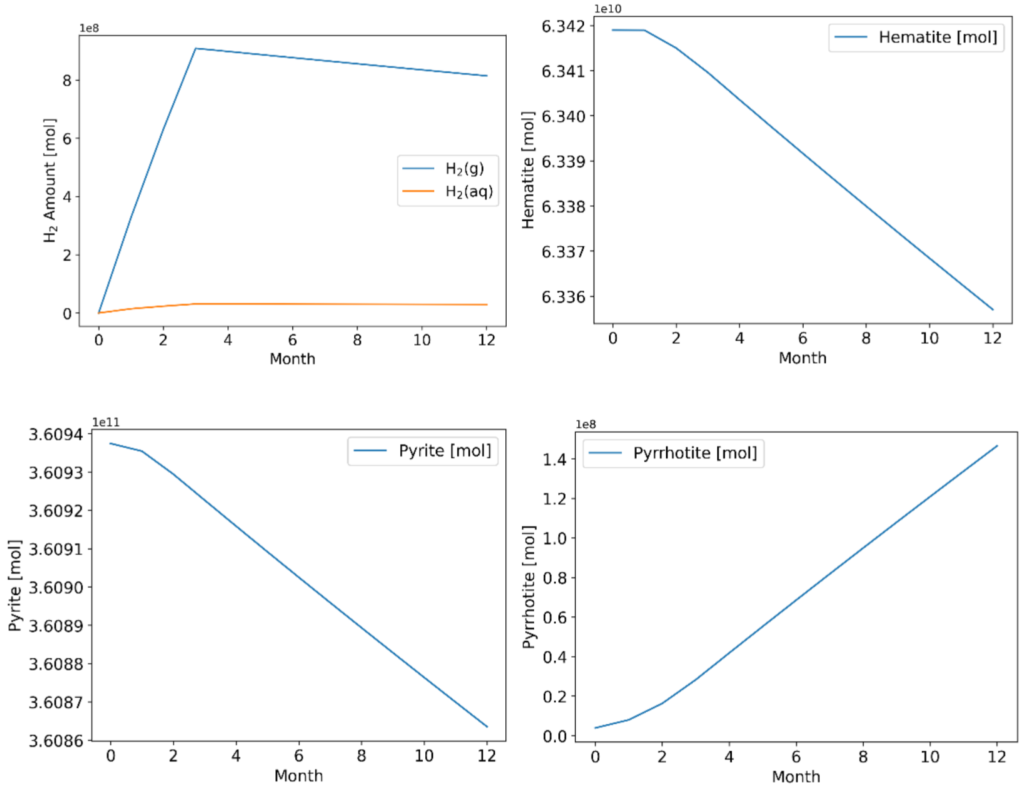

Figure 7—Variation of the simulated amount of H2 (top left), hematite (top right), pyrite (bottom left), and pyrrhotite (bottom right) in the storage aquifer during the first 12 months of storage. For the H2 storage case, H2(g) loss due to dissolution is close to 19%.

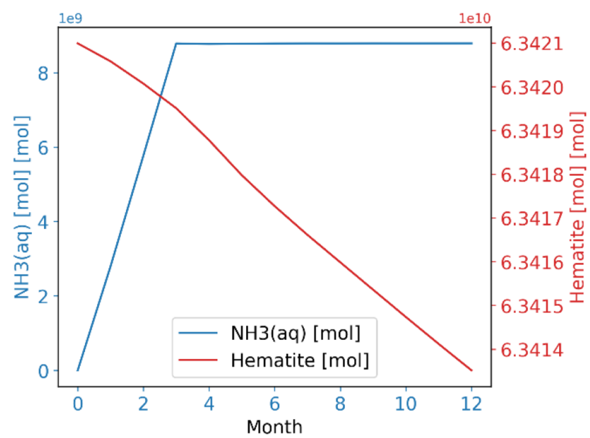

Figure 8—Variation of the simulated amount of NH3 in the storage aquifer during the first 12 months of storage. For the NH3 storage case, hematite reduction is the major source of loss.

Conclusions

Based on reactive transport simulations, the following conclusions can be drawn:

- Iron (III)- and sulfur reductions are primarily responsible for the loss of H2 gas in the subsurface environment.
- Clay and silicate minerals exhibit minimal reactivity with H2 gas, suggesting that sandstone-rich reservoirs could be viable for hydrogen storage.

- Given its distinct properties, such as energy density, transportation cost, physical trapping efficiency, and operational safety, ammonia (NH_3) warrants further exploration as an alternative hydrogen carrier for subsurface storage.
- Iron(III) reduction is primarily responsible for the loss of NH3 gas in the subsurface environment. However, NH_3 is chemically much more stable than H2 in the subsurface.
- When accounting for losses attributable to geochemical reactions, the storage capacity of NH_3 as a hydrogen carrier could potentially be around 16 times more compared to that of H2 gas for the synthetic case studied.

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